

Ritwick Banerjee

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Summary

Data Scientist with 4+ years of experience in big data analysis, statistical inference, machine learning, and NLP research. Skilled in Data Science, Python, SQL, and machine learning, with a proven track record of improving efficiency, boosting outreach, plugging revenue leakages for clients, delivering impact of over \$100,000 in revenue.

Experience

Graduate Research Assistant, University of Minnesota – Minneapolis, MN

Jan 2024 – Present

- Analyzed over 150K student records using Python and Power BI to identify student achievement gaps, employed normalizing techniques to identify factors impacting year-on-year GPA & enrollment, leading to a 20% increase in targeted academic support for underperforming student groups.
- Conducted statistical analyses (Cramer's V, Point Biserial correlation, Chi-square tests) to identify key success factors, leading to evidence-based educational interventions.
- Integrated causal discovery techniques to explore causal relationships in educational survey data to uncover unknown insights.

Data Analyst, University of Minnesota – Minneapolis, MN

Sep 2023 – Present

- Developed a streamlined data pipeline, increasing data collection efficiency by 40% for the Mobile Health Initiative, which improved outreach and decision-making processes.
- Boosted marketing campaign effectiveness by 32% through applied statistical methods (e.g. Chi-square, ANOVA, Lasso regression) to analyze patient response data, leading to more targeted and impactful campaigns.
- Collaborated with cross-functional teams and developed interactive dashboards on the patient interactions, demographics, and screening results to enable data-driven decision making.

Associate Consultant, Team Lead, KPMG – Gurugram, India

Jan 2021 – Aug 2023

- Designed and implemented a revenue assurance framework with a team of five, resulting in a daily resource hours saving of 16 hours
- Developed interactive dashboards for monitoring roaming CDRs and customer rates, optimizing revenue assurance and saving 24 hours of team bandwidth per day.
- Conducted analyses to avert revenue leakage during critical product launches and the on-boarding of new partners safeguarding up to \$ 70,000 in revenue.

Education

Master of Science in Data Science

Sept 2023 – May 2025

University of Minnesota • Minneapolis, MN, USA • **GPA:** 3.8/4.0

- **Coursework:** Artificial Intelligence, NLP, Applied Multivariate & Regression analysis, Machine Learning, Causal Discovery

Bachelor of Technology in Computer Science

July 2017 – June 2021

KIIT University • Bhubaneswar, OD, India • **GPA:** 8.6/10.0

Projects

Resample or Not?

[Github Link](#)

- Investigated the effects of different resampling methods on both score and constraint based causal discovery algorithms' stability for both ensemble models and calibration of the models. Bootstrap and 90% sub-sampling are the best performers.

Causal effects of Mood: Depression, Mania and more

[Github Link](#)

- Explored the complex causal relationships between manic and depressive symptoms across various mood states, including mania, depression, mixed states, and euthymia using a hybrid(score based and constraint based) causal discovery algorithm.

Patient Pre-screening Chatbot

[Github Link](#)

- Developed a LLM powered chatbot using RAG to pre-screen patients based on their symptoms and medical history for health services.

Technologies

Frameworks, Tools and Languages: C++, C, Python, SQL, R, Causal Inference, Tetrad, Deep Learning, Docker, Exploratory Analysis, Git, Github, GCP, Bigquery, HuggingFace, Langchain, NLP, NLTK, NumPy, Pandas, PowerBI, Tableau, Qlik sense, Qlik view, Jira, Kubernetes, Pytorch, Seaborn, Scikit-learn, Anova, F, T, chi-square tests, Transformers, Decision Trees, K-means, SVMs, Random forests, AdaGrad, Adaboost, XGBoost

Soft Skills: Adaptability, Cross-functional Collaboration, Interpersonal Communication, Problem Solving, Project Management